

# Vishal Kumar Mathuri

BLOCKCHAIN ENGINEER | SOLIDITY / EVM | RUST / SOLANA

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## SUMMARY

Blockchain Engineer with hands-on professional experience building **Solidity smart contracts, EVM protocols, DeFi systems, tokenized applications**, and full-stack Web3 products — including **50+ smart contracts** and **10+ full-stack blockchain applications** shipped to date. Strong focus on **secure and testable smart-contract engineering** using Solidity, Foundry, Hardhat, OpenZeppelin, and Ethers.js, with full-stack experience across React.js, Node.js, and REST APIs.

Expanding into **Rust/Solana and cross-chain protocol engineering**, with a portfolio centered on **production-oriented protocol design** rather than a collection of tutorial projects.

## TECHNICAL SKILLS

**Smart Contracts:** Solidity, ERC-20, ERC-721, access control, staking, governance, vesting, custom errors, events

**EVM / Blockchain:** Ethereum, EVM, Polygon, Ethereum Sepolia, gas optimization, contract verification, IPFS

**Security / Testing:** OpenZeppelin, ReentrancyGuard, AccessControl, Ownable, Pausable, unit testing, fuzz testing, invariant testing

**Web3 / Backend:** Ethers.js, Web3.js, React.js, Node.js, Express.js, REST APIs, MetaMask

**Rust / Solana:** Rust, Solana, Anchor, PDAs, TypeScript — actively developing production-oriented Solana projects

**Tools:** Foundry, Hardhat, Remix, Git, GitHub, VS Code, Vercel, Render, IPFS/Pinata

## PROFESSIONAL EXPERIENCE

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| <b>Blockchain Developer</b><br>Freelance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Jul 2025 - Present</b>                   |
| <ul style="list-style-type: none"><li>Developed and deployed <b>50+ Solidity smart contracts</b> implementing ERC-20, ERC-721, staking, governance, and marketplace functionality across NFT, fundraising, identity, and access-control applications.</li><li>Built <b>10+ full-stack Web3 applications</b> integrating smart contracts with <b>React.js frontends, Node.js backend services, REST APIs, wallets, and EVM networks</b>.</li><li>Applied <b>OpenZeppelin security primitives</b> and secure design patterns including reentrancy protection and role-based access control, reducing security risk across deployed contracts.</li><li>Worked across contract implementation, testing, deployment, frontend integration, debugging, and production issue resolution.</li></ul>                                                                                                                                                                                                                                                                                     |                                             |
| <b>Senior Content Developer</b><br>Think & Learn Pvt Ltd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Jul 2019 - Mar 2023</b><br>Bengaluru, IN |
| <ul style="list-style-type: none"><li>Managed scheduling operations for 200+ student batches across multiple grade levels with <b>100% operational accuracy</b> and enhanced workflow efficiency.</li><li>Improved an automated concept-tagging workflow to <b>95% classification accuracy</b> through systematic validation and quality checks, boosting system efficiency and reducing errors.</li><li>Crafted and executed structured Learning Journeys for academic programs utilizing digital learning platforms, resulting in a <b>96% success rate</b>.</li><li>Designed and curated interactive educational content, mock tests, and learning modules for Byju's TV and BHELP programs, increasing user satisfaction and engagement.</li><li>Resolved student support and academic operations requests using Salesforce with <b>98% resolution accuracy</b> within 24 hours, ensuring uninterrupted course delivery.</li><li>Collaborated with academic operations, HR, and support teams while maintaining strong cross-team communication and coordination.</li></ul> |                                             |

## FLAGSHIP BLOCKCHAIN PROJECTS

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| <b>1. Production-Grade EVM Lending Protocol</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
| <b>Tech Stack:</b> Solidity • Foundry • Chainlink • OpenZeppelin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| <b>Links:</b> <a href="#">GitHub</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| <ul style="list-style-type: none"><li>Engineered a modular, overcollateralized EVM lending protocol supporting multi-asset supply, withdrawal, borrowing, repayment, interest accrual, and liquidation, with responsibilities separated across <b>LendingPool, ReserveManager, RiskManager</b>, and configurable interest-rate modules.</li><li>Implemented <b>index-based scaled accounting and risk management</b>, including utilization-driven borrow/supply rates, collateral-factor borrowing limits, health-factor solvency checks, liquidation thresholds/bonuses, multi-decimal asset normalization, and rounding-aware debt/collateral accounting.</li><li>Built a <b>Chainlink-based oracle layer</b> with defensive price validation, including stale-price protection, incomplete/stale round detection, positive-price validation, future-timestamp checks, configurable feed heartbeats, and an oracle abstraction enabling deterministic mock testing.</li><li>Validated the protocol with an extensive <b>Foundry test suite spanning unit, integration, fuzz, and stateful invariant testing</b>, plus failure-path and security suites; deployed and source-verified the contracts on Ethereum Sepolia and executed an end-to-end supply → borrow → repay → withdraw flow.</li></ul> |  |
| <b>2. NFT Marketplace</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| <b>Tech Stack:</b> Solidity • ERC-721 • OpenZeppelin • React.js • Ethers.js • IPFS/Pinata                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| <b>Links:</b> <a href="#">GitHub</a> • <a href="#">Live (Vercel)</a> • <a href="#">API (Render)</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
| <ul style="list-style-type: none"><li>Engineered a full-stack NFT marketplace supporting ERC-721 minting, listing, purchasing, and ownership transfers through Solidity smart contracts, with a React.js interface for end-to-end marketplace interactions.</li><li>Integrated <b>Ethers.js and MetaMask</b> for wallet-based Web3 workflows, enabling users to connect accounts, sign transactions, mint NFTs, list assets, and execute purchases directly against deployed smart contracts.</li><li>Implemented decentralized NFT metadata storage using <b>IPFS/Pinata</b>, linking token metadata and digital assets to ERC-721 tokens while using OpenZeppelin contracts for standardized NFT functionality and ownership controls.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |

- Delivered a publicly accessible full-stack application, deploying the React frontend on Vercel and backend services on Render to provide a live, testable Web3 application beyond standalone smart contracts.

3. Solana Crowdfunding Protocol

Tech Stack: Rust • Solana • Anchor • PDAs • TypeScript • Wallet Adapter

Status: In Development    Links: [GitHub](#)

- Building a Solana crowdfunding protocol in Rust and Anchor for campaign creation, SOL contributions, funding-goal tracking, deadline enforcement, creator withdrawals, and contributor refunds.
- Designing **PDA-based campaign and protocol state**, using deterministic account derivation to associate campaign data and protocol-controlled accounts with campaign creators without relying on externally generated private keys.
- Implementing Anchor account constraints and authorization rules to validate campaign ownership, signer permissions, contribution state, withdrawal eligibility, and refund conditions at the program level.
- Developing the TypeScript/Wallet Adapter integration layer for wallet-connected campaign interactions while structuring tests around successful funding, failed campaigns, unauthorized withdrawals, deadline boundaries, and refund behavior.

4. Cross-Chain Messaging Protocol — EVM ↔ Rust

Tech Stack: Solidity • Rust • EVM • Cross-Chain Messaging • Relay • Foundry

Status: In Development    Links: [GitHub](#)

- Building a cross-chain messaging protocol connecting EVM contracts with a second blockchain environment, using a relay architecture to transport and execute authenticated messages across otherwise isolated execution environments.
- Designing deterministic message identities, source/destination metadata, nonces, and replay protection to ensure cross-chain messages are uniquely tracked and cannot be executed multiple times on the destination chain.
- Implementing authorization and message-verification boundaries across Solidity and Rust components, with explicit handling for message lifecycle states, failed execution, retries, and invalid or unauthorized messages.
- Designing protocol security controls for cross-chain failure scenarios, including relay authorization, emergency controls, duplicate-message protection, and defensive validation before destination-chain execution.

INTERNSHIPS

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| <b>Blockchain Developer</b><br>EtherAuthority                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Mar 2026 - May 2026</b><br>Remote |
| <ul style="list-style-type: none"><li>• Built a Telegram Mini App featuring multiple in-app mini-games, including Coinflip, Find the King (a dice-based game), and Wheel, using the Telegram Web App platform.</li><li>• Designed and implemented game logic, betting flow, and result computation for each game mode.</li><li>• Collaborated with the team on frontend/backend integration and testing to ship engaging, low-latency gameplay experiences.</li></ul> |                                      |
| <b>Blockchain Developer</b><br>Leafny Web Pack                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>May 2025 - Jul 2025</b><br>Remote |
| <ul style="list-style-type: none"><li>• Assisted in building Web3 dApps integrating smart contracts with frontend and backend systems.</li><li>• Worked with ERC-20 and ERC-721 token standards and participated in smart-contract security testing and reviews.</li></ul>                                                                                                                                                                                            |                                      |
| <b>Blockchain Developer</b><br>Estatos                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Jan 2025 - Feb 2025</b><br>Remote |
| <ul style="list-style-type: none"><li>• Developed and optimized smart contracts for blockchain operations.</li><li>• Worked on tokenization and DeFi protocols, contributing to secure and efficient smart-contract deployments.</li><li>• Collaborated with the tech team to enhance solutions while demonstrating problem-solving and a growth mindset.</li></ul>                                                                                                   |                                      |
| <b>Solidity Developer</b><br>Nexus Network                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Jul 2024 - Dec 2024</b><br>Remote |
| <ul style="list-style-type: none"><li>• Developed bridge contracts for relay-based cross-chain transactions.</li><li>• Strengthened smart-contract security and optimization practices.</li><li>• Actively participated in team collaborations and implemented feedback for continuous improvement.</li></ul>                                                                                                                                                         |                                      |

EDUCATION

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| <b>Post Graduate Diploma in Blockchain</b><br>IIIT Bangalore & upGrad                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Mar 2023 - Mar 2024</b><br>Bengaluru, IN |
| <ul style="list-style-type: none"><li>• Completed advanced coursework covering JavaScript, Node.js, Ethereum smart-contract development, and Hyperledger Fabric.</li><li>• Developed Ethereum-based smart-contract architectures for decentralized applications including DeFi systems.</li><li>• Designed system architecture for the UPSTAC healthcare application, focusing on scalable and secure blockchain integration.</li><li>• Explored blockchain-based KYC systems, Hyperledger Fabric network setup, and cross-chain frameworks.</li></ul> |                                             |
| <b>Dual Degree (B.Tech + M.Tech), Energy Science &amp; Engineering</b><br>Indian Institute of Technology Bombay (IIT Bombay)                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Jul 2014 - Jul 2019</b><br>Mumbai, IN    |
| <ul style="list-style-type: none"><li>• Secured 61%.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                             |

ADDITIONAL INFORMATION

- **Languages:** Fluent in Hindi (Native) and English (Professional)
- **Interests:** Reading and Cycling